

VAISHNAVI ANNAM

Mumbai, India



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LinkedIn



GitHub

Portfolio

Summary

Information Technology undergraduate pursuing Honours in Artificial Intelligence and Machine Learning with a CGPA of **9.5/10**. Skilled in developing machine learning solutions using Python, NumPy, Pandas, and Scikit-learn, with hands-on experience in Random Forest Classification, K-Means Clustering, and Recommendation Systems. Passionate about applying data-driven techniques to solve real-world challenges in urban analytics, customer segmentation, and intelligent decision-making.ss

Education

Atharva College of Engineering, Mumbai

Bachelor of Engineering in Information Technology

Honours in Artificial Intelligence and Machine Learning

Expected 2027

CGPA: 9.5/10.0

- Ranked among the top-performing students in the department.
- Consistently maintained academic excellence throughout the degree program.

Technical Skills

Programming Languages: Python, Java, JavaScript, SQL

Machine Learning & Data Analysis: NumPy, Pandas, Scikit-learn, K-Means Clustering, Data Preprocessing, Exploratory Data Analysis, Data Visualization

Core Subjects: Data Structures & Algorithms, Probability & Statistics, Linear Algebra, Machine Learning Fundamentals, DBMS, OOP, Operating Systems

Web Development: React.js, Django, Node.js, REST APIs, HTML5, CSS3, Bootstrap, Tailwind CSS

Databases & Tools: MySQL, PostgreSQL, Git, GitHub, VS Code

Machine Learning Projects

Customer Segmentation using K-Means Clustering

Python, NumPy, Pandas, Scikit-learn, Matplotlib

- Developed a machine learning model to segment customers based on behavioral and purchasing patterns using K-Means clustering.
- Performed data preprocessing, feature selection, and exploratory data analysis to prepare customer data for clustering.
- Applied the Elbow Method to determine the optimal number of customer segments.
- Visualized clusters to support data-driven marketing and customer targeting strategies.

GitHub: github.com/Vaishnavi688/Customer_Segmentation_Kmeans

Urban Heat Island Analysis of Mumbai

Google Earth Engine, Python, Random Forest, Linear Regression

- Analyzed Landsat satellite imagery (2000–2025) to study land-use/land-cover changes across Mumbai using Random Forest classification.
- Calculated NDVI, NDBI, NDWI, and Land Surface Temperature (LST) to assess the impact of urbanization on temperature trends.
- Performed correlation analysis and identified a strong relationship between built-up areas and increased surface temperatures.
- Predicted Mumbai's 2030 land-use patterns using Linear Regression and visualized trends through Python-based data analysis.

GitHub: github.com/Vaishnavi688/mumbai-urban-heat-analysis

Movie Recommendation System

Python, Pandas, Scikit-learn, KNN

- Developed a movie recommendation system using collaborative filtering and K-Nearest Neighbors (KNN).
- Built a user-item interaction matrix from the MovieLens dataset and computed user similarity using cosine distance.
- Generated personalized movie recommendations based on user rating patterns and preferences.
- Performed exploratory data analysis and visualized rating distributions and movie popularity trends.

GitHub: github.com/Vaishnavi688/Movie-recommendation-system

Projects

NutriScan – AI-Powered Food Label Analyzer

React.js, Node.js, PostgreSQL, OCR

- Developed an OCR-powered food label analysis platform to extract ingredients and nutritional details from product labels.
- Implemented text-processing logic to identify ingredients, allergens, and dietary concerns from extracted label data.
- Built a mobile-first interface enabling users to scan food labels and receive structured nutritional insights.

SakhiCare – Online Women’s Healthcare System

Django, Python, MySQL

- Developed a healthcare platform supporting menstrual cycle tracking, pregnancy monitoring, and doctor appointment scheduling.
- Implemented secure authentication, personalized dashboards, emergency helpline services, and healthcare resources.

Leadership Experience

Training & Placement Student Coordinator

- Coordinated placement-related activities between students, faculty coordinators, and recruiters.
- Assisted in organizing placement drives, pre-placement talks, registrations, documentation, and student communication.

Problem Solving

- Solved 50+ problems on LeetCode covering arrays, 2D arrays, searching, and fundamental problem-solving techniques.
- Strengthening Data Structures and Algorithms with focus on time complexity, optimization, and coding assessments.

Certifications and Achievements

- Cybersecurity Hands-on Workshop – Google Developer Groups (2025)
- Java Programming – Great Learning (2024)
- Maintained a CGPA of 9.5/10 and ranked among the top-performing students in the department.
- Developed ML, AI-enabled, and full-stack applications across customer analytics, food safety, and healthcare domains.